

Rizwan Ali Shah

AI Research and Development Engineer | Computer Vision | Industrial AI

 rizwanali@cbnu.ac.kr  South Korea  RizwanAliQau  LinkedIn

Professional Summary

AI Research and Development Engineer specialising in deep learning-based anomaly detection, industrial computer vision, self-supervised learning, synthetic anomaly generation, and smart manufacturing. Experienced in developing robust and scalable inspection systems, creating synthetic and real-world datasets, and integrating vision solutions into automated manufacturing workflows. Published in *IEEE Access*, *Image and Vision Computing*, and peer-reviewed international conferences.

Research Highlights

3
SCI journal publications
4
Working manuscripts

5
Peer-reviewed conference papers
5
Industrial AI projects

2
Journal papers under review
3+
Years teaching and industrial AI experience

Professional Experience

AI Research and Development Engineer

MSIS Robotics – Full-time

Sep. 2025–Present
South Korea, On-site

- Develop and optimise AI and deep learning models for industrial vision, defect detection, and quality control.
- Design robust image-processing algorithms for variable and challenging manufacturing conditions.
- Apply self-supervised learning and synthetic anomaly generation to reduce reliance on scarce defective samples.
- Build and curate large-scale labelled datasets combining synthetic and real industrial images.
- Collaborate with cross-disciplinary teams to integrate vision systems into automated manufacturing workflows.
- Conduct experimental evaluations to improve model robustness, adaptability, accuracy, and operational efficiency.

Lecturer, Department of Electronics

Quaid-i-Azam University

2018–2021
Islamabad, Pakistan

- Taught Computer Vision, Digital Image Processing, Computer Programming, and Applied Electronics.
- Designed lectures, laboratory exercises, and assessments using Python, MATLAB, and image-processing tools.
- Supervised undergraduate research and capstone projects in computer vision, machine learning, and electronics.

Education

Ph.D. in Electronics Engineering

Chungbuk National University

2021–2025
Cheongju, South Korea

Dissertation: Self-Supervised Deep Learning Technique for Multi-Class Industrial Anomaly Detection Based on Synthetic Anomaly Insertion.

Research areas: Computer vision, deep learning, industrial anomaly detection, and image processing.

M.Phil. in Electronics

Quaid-i-Azam University

2015–2017
Islamabad, Pakistan

Concentration in digital image processing and pattern recognition.

Selected Industrial AI Projects

AI Robot Inspection for Connector Anomalies

2024–2025

AI-enabled robotic vision pipeline for real-time connector localisation and anomaly inspection.

Defect Inspection of FPCB for Electric Vehicle Batteries

2023–2024

High-precision defect detection and localisation for flexible printed circuit boards.

Malfunction Detection in Injection Molding

2023–2024

Self-supervised inspection using detector models, template matching, and colour pseudo-anomalies.

Anomaly Detection in Connector Component Assembly

2023

Deep learning and self-supervised feature learning for connector assembly inspection.

Anomaly Detection in Electric Vehicle Battery Assembly

2022

Generative anomaly modelling and synthetic defect generation for limited defective samples.

 Journal Publications

2025	Self-Supervised Image Anomaly Detection Through Diverse Pseudo Anomaly Insertion <i>IEEE Access</i>	First Author
2025	Patch-Shuffled Superpixel Synthetic Anomaly Insertion in Autoencoder Foreground Extracted Regions for Improved Visual Anomaly Detection <i>IEEE Access</i>	Co-Author
2023	Two-Stage Coarse-to-Fine Image Anomaly Segmentation and Detection Model <i>Image and Vision Computing, Elsevier</i>	First Author

 Journal Papers Under Review

2026	RAUDI: A Dual-Head Framework Unifying Unsupervised and Supervised Reconstruction for Industrial Anomaly Detection <i>Image and Vision Computing</i>	Co-Author
2026	CSR: Continual Subspace Refinement for One-Shot Industrial Anomaly Detection <i>IEEE Transactions on Industrial Informatics</i>	Co-Author

 Peer-Reviewed Conference Papers

2025	Multi-Class Segmentation Network for Robust Class-Specific Anomaly Detection <i>Summer Conference of the Korea Electronic Engineering Association</i>	Author
2024	Patch-Oriented Anomaly Detection and Segmentation Model <i>Korean Institute of Communications and Information Sciences AI Conference</i>	Author
2023	Advanced Color Pseudo Anomaly to Enhance Learning of Convolutional Neural Network Models <i>IEEE International Conference on Consumer Electronics-Asia</i>	Author
2022	Enhancement of Anomaly Detection Using Two-Stage Anomaly Segmentation Model <i>8th International Conference on Next Generation Computing</i>	Author
2021	Improving Performance of CNN-Based Vehicle Detection and Tracking by Median Algorithm <i>IEEE International Conference on Consumer Electronics-Asia</i>	Author

 Working Manuscripts

- **A Multi-Signal Dual-Decoder Framework for Unsupervised Industrial Anomaly Detection** – Co-author.
- **Federated Learning with Orthogonally Enhanced Aligned Client Self-Training** – Co-author.
- **Multi-Channel Agnostic Self-Supervised Multi-Class Anomaly Detection for Industrial Datasets** – First author.
- **Exploiting Geometric Symmetries to Enhance Structural Anomaly Segmentation in Industrial Vision** – First author.

 Technical Skills

Programming	Python, MATLAB
Frameworks	PyTorch, TensorFlow, OpenCV
AI and Vision	Deep learning, self-supervised learning, anomaly detection, image segmentation, synthetic data generation, CNNs, Vision Transformers
Tools	Linux, Git, LaTeX

 Honors and Awards

2024	MSISLAB Scholarship for industrial-academic technology excellence.
2022	Best Poster Award, 8th International Conference on Next Generation Computing.
2021–2025	Brain Korea 21 Graduate Fellowship, National Research Foundation of Korea.

Academic References

Prof. HyungWon Kim
Professor
Department of Electronics Engineering
Chungbuk National University
Cheongju, South Korea
hwkim@cbnu.ac.kr

Dr. M. Naeem Ali Bhatti
Associate Professor
Department of Electronics
Quaid-i-Azam University
Islamabad, Pakistan
nbhatti@qau.edu.pk